Math 113– Lecture 02

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1 Groups

A group is a binary algebraic structure $(G, *)$ such that

- $*$ is associative $\forall a, b, c \in G, a * (b * c) = (a * b) * c$
- $\exists e \in G$ such that e is an identity element $e * a = a * e = a$
- $\forall a \in G, \exists a^{-1} \in G$ (each element has an inverse)
- In this class, assume it's a two sided inverse, namely $aa^{-1} = a^{-1}a = e$

Definition 1.1 (Abelian Group)**.** A group G is abelian if $*$ is commutative $(\forall a, b \in G, a * b = b * a.)$

2 Definitions and notation

Theorem 2.1 (Cancellation Law for Groups) *For a group* $(G, *)$ *and* $a, b, c \in G$*, we have*

$$a * b = a * c \iff b = c$$

Proof.

$$\begin{aligned} a * b = a * ca^{-1} * (a * b) &= a^{-1} * (a * c) \\ (a^{-1} * a) * b &= (a^{-1} * a) * c \\ e * b &= e * c \to b = c \end{aligned}$$

□

Theorem 2.2 (Uniqueness) *For a group* $(G, *)$ *and* $a, b \in G$*, then*

$$a * x = b \text{ has a unique solution } x \in G$$

Proof. Existence: x = $a^{-1}b$ is a solution.

Uniqueness: Suppose for sake of contradiction, $\exists y \in G$ such that $a * x = b = a * y$

However, by the cancellation law $a * x = a * y \iff x = y$ a contradiction. □

Theorem 2.3 (Uniqueness of Inverses) *For $a \in G$ of a group $(G, *)$, it has a unique inverse a^{-1}.*

Proof. Any inverse is a solution to $a * x = e$, and by the uniqueness theorem, it has a unique solution ☐

Corollary 2.4 (Corollary to Previous Theorem) *For $a, b \in G$ of a group $(G, *)$, we have*

$$(a * b)^{-1} = b^{-1}a^{-1}$$

Proof. We check it:

$$\begin{aligned}(a * b) * (b^{-1} * a^{-1}) &= a * (b * b^{-1}) * a^{-1} \\ &= a * e * a^{-1} \\ &= a * a^{-1} \\ &= e\end{aligned}$$

☐

Definition 2.5 (First definition)**.** Write the definition and one sentence explaining why it matters.

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example)**.** Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: